Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-01-C01-BD1

Brain dump 1 · foundational retrieval: Blood pressure categories and why the kidney sets pressure

A competent response contains

· Normal is below 120/80 mmHg, elevated is 120 to 129 systolic with diastolic below 80, stage 1 is 130 to 139 systolic or 80 to 89 diastolic, stage 2 is 140 or higher systolic or 90 or higher diastolic.

· When the systolic and diastolic values fall in different categories the higher category applies, so 128/86 mmHg is stage 1.

· Diagnosis rests on the average of repeated readings on two or more occasions, ideally confirmed outside the office.

· Mean arterial pressure equals cardiac output multiplied by systemic vascular resistance; in established primary hypertension output is usually normal and resistance is raised.

· Cardiovascular risk rises continuously from about 115 mmHg systolic, so any threshold is a chosen cut point.

· Baroreceptors defend pressure over minutes but reset within a day or two, so they cannot set long-term pressure.

· The kidney sets long-term pressure: a rise in arterial pressure increases sodium and water excretion, which is pressure natriuresis, and volume falls until sodium output again equals sodium intake.

· Because the gain of this feedback is very high, pressure can stay high only if the renal pressure to sodium excretion relationship shifts to the right, so a higher pressure is needed to excrete the daily salt load.

· Every chronic hypertension, primary or secondary, involves such a shift in pressure natriuresis.

Distinctions that matter

· Short-term pressure control is neural and resets; long-term control is renal and does not.

· A raised resistance is the usual finding in established disease, which is not the same as saying resistance started the process.

Common misconceptions

· Treating 140/90 mmHg as the threshold for hypertension under the scheme used here. Stage 1 begins at 130/80 mmHg under the 2017 ACC/AHA guideline.

· Saying baroreceptors keep pressure high in chronic hypertension. They reset to the prevailing pressure within a day or two.

Where it is taught: Section C01, how blood pressure is classified and why pressure stays high of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C01-BD2

Brain dump 2 · integration and application: Office readings, night-time pressure, and two inherited salt channels

A competent response contains

· Patient one has white coat hypertension, high only in the office from an alerting sympathetic response, so the clinic category of stage 1 overstates the true pressure.

· Patient two has masked hypertension, high only outside the office, so a normal clinic reading understates the true pressure.

· Patient three has loss of nocturnal dipping; pressure normally falls about 10 to 20% during sleep, and loss of the fall suggests sleep apnea, kidney disease, or autonomic dysfunction.

· Liddle syndrome: a gain-of-function mutation keeps the epithelial sodium channel of the collecting duct in the membrane, so sodium is reabsorbed regardless of aldosterone.

· Apparent mineralocorticoid excess: deficiency of 11-beta-hydroxysteroid dehydrogenase type 2 lets cortisol occupy the mineralocorticoid receptor.

· Glycyrrhetinic acid in licorice inhibits the same enzyme and reproduces the picture.

· In each one, sodium reabsorption through the epithelial sodium channel makes the lumen more negative, which drives potassium secretion by principal cells and hydrogen ion secretion by alpha-intercalated cells, giving hypokalemia and metabolic alkalosis.

· Volume expansion suppresses renin, and because the drive is not aldosterone itself, aldosterone is also low, so both are low with early-onset hypertension.

· Risk factors acting partly through sympathetic outflow: obesity through leptin, heavy alcohol intake, physical inactivity, and sleep apnea.

· Risk factors acting mainly through volume and renal sodium handling: high sodium with low potassium intake, hyperinsulinemia of obesity, diabetes, and chronic kidney disease.

· Older age acts through aortic stiffening, which raises systolic and pulse pressure and lowers diastolic pressure, giving isolated systolic hypertension.

Distinctions that matter

· White coat hypertension is high in the office only; masked hypertension is high outside it only. The out-of-office reading settles which one it is.

· Low renin with high aldosterone points to the adrenal cortex; low renin with low aldosterone points to a mineralocorticoid effect from something other than aldosterone.

Common misconceptions

· Expecting aldosterone to be high whenever a patient retains sodium and wastes potassium. In Liddle syndrome and apparent mineralocorticoid excess the receptor or channel is active without aldosterone.

· Assigning a stage from a single office reading. The category comes from averaged repeated readings, ideally confirmed outside the office.

Where it is taught: Section C01, paragraphs on measurement and on polygenic and monogenic hypertension, and the risk factor table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C02-BD1

Brain dump 1 · foundational retrieval: Build the secondary hypertension table from memory

A competent response contains

· Renal artery stenosis: abdominal bruit, hypokalemia, small kidney on the affected side, and a fall in glomerular filtration rate when angiotensin II is blocked; renin high, aldosterone high.

· Primary aldosteronism: hypokalemia, metabolic alkalosis, no edema; renin low, aldosterone high.

· Cushing syndrome, licorice, and Liddle syndrome: cushingoid habitus, licorice intake, or early onset with hypokalemia; renin low, aldosterone low or normal.

· Pheochromocytoma: episodic headache, sweating, and palpitations with high plasma or urine metanephrines; renin and aldosterone are not the primary abnormality.

· Coarctation of the aorta: arm hypertension with weak and delayed femoral pulses and rib notching; renin may be raised.

· Obstructive sleep apnea: snoring, daytime sleepiness, loss of nocturnal dipping; renin variable.

· Atherosclerosis causes most renal artery stenosis, in older adults with vascular disease, and narrows the ostium or proximal artery.

· Fibromuscular dysplasia is a noninflammatory, usually medial thickening in women about 15 to 50 years old, affecting the mid to distal artery with a string of beads appearance, and can involve the carotid and vertebral arteries.

· In pheochromocytoma, alpha-1 receptors raise systemic vascular resistance and beta-1 receptors raise cardiac output.

· Secondary hypertension is about 5 to 10% of all cases.

· Look for a cause when hypertension begins before about age 30, begins abruptly, is severe or resistant, or comes with an unexpected finding such as hypokalemia.

· Chronic kidney disease of any cause is a common secondary cause, because fewer nephrons must excrete the same salt load.

Distinctions that matter

· Both renin and aldosterone high points to renal hypoperfusion; low renin with high aldosterone points to the adrenal cortex; both low points to a mineralocorticoid effect from something other than aldosterone.

· Atherosclerotic and fibromuscular renal artery disease differ in age, sex, and the part of the artery involved, so keep the two profiles apart.

Common misconceptions

· Expecting every secondary cause to change renin and aldosterone. In pheochromocytoma the catecholamines are the abnormality and those two values are not the test.

· Assuming hypokalemia means primary aldosteronism. Renal artery stenosis, Cushing syndrome, licorice, and Liddle syndrome all produce it.

Where it is taught: Section C02, opening paragraph, renal artery stenosis, adrenal causes, and the secondary causes table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C02-BD2

Brain dump 2 · integration and application: Reason from the laboratory pattern back to the organ

A competent response contains

· Patient A has renal artery stenosis. The narrowed artery lowers afferent arteriolar pressure, so juxtaglomerular cells release renin as though the whole body were underperfused, and angiotensin II then drives aldosterone, so both are high.

· Patient B has primary aldosteronism, from bilateral adrenal hyperplasia or an aldosterone-producing adenoma. Autonomous aldosterone expands volume, and volume expansion suppresses renin, so renin is low while aldosterone is high.

· Patient B's tubular steps: aldosterone opens more epithelial sodium channels in principal cells, the more negative lumen drives potassium secretion, and alpha-intercalated cells secrete more hydrogen ion, producing hypokalemia with metabolic alkalosis.

· Patient B has no edema because rising pressure and atrial natriuretic peptide restore sodium balance within days, which is aldosterone escape, while potassium wasting continues; many such patients are in fact normokalemic.

· Patient C has a mineralocorticoid effect from cortisol. Glycyrrhetinic acid in licorice inhibits 11-beta-hydroxysteroid dehydrogenase type 2, so cortisol occupies the mineralocorticoid receptor; volume expansion suppresses renin and aldosterone is not the driver, so both are low.

· Patient D has a pheochromocytoma, a catecholamine-secreting chromaffin cell tumor of the adrenal medulla, called a paraganglioma when extra-adrenal.

· Metanephrines stay high between attacks because catechol-O-methyltransferase within the tumor forms them continuously, while catecholamine release is episodic.

· An epinephrine-secreting tumor is almost always adrenal because cortisol from the adrenal cortex induces phenylethanolamine N-methyltransferase, the enzyme that converts norepinephrine to epinephrine.

· Hereditary pheochromocytoma occurs in MEN 2 with RET mutations, von Hippel-Lindau disease, and neurofibromatosis type 1.

· Filtration in a stenotic kidney depends on angiotensin II constricting the efferent arteriole, so blocking angiotensin II dilates that arteriole, drops glomerular capillary pressure, and can sharply lower the glomerular filtration rate.

· Bilateral renal artery disease can cause recurrent flash pulmonary edema.

Distinctions that matter

· Renin separates renal artery stenosis from primary aldosteronism, because one starts with renal hypoperfusion and the other starts with volume expansion.

· Aldosterone separates adrenal cortical excess from a cortisol or channel effect at the mineralocorticoid receptor, since both suppress renin but only one raises aldosterone.

Common misconceptions

· Expecting edema in primary aldosteronism. Aldosterone escape restores sodium balance, so pressure rises without visible fluid retention.

· Reading a normal potassium as evidence against primary aldosteronism. Many patients are normokalemic.

Where it is taught: Section C02, renal artery stenosis, adrenal causes, and the secondary causes table caption of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C02-CP01

Comparison table: Renal artery stenosis against primary aldosteronism

A competent response contains

· Site: renal artery stenosis is a narrowed main renal artery, from atherosclerosis in older adults or fibromuscular dysplasia in younger women; primary aldosteronism is in the adrenal cortex, from bilateral hyperplasia or an aldosterone-producing adenoma.

· Renin: high in renal artery stenosis because reduced afferent arteriolar pressure makes juxtaglomerular cells release renin; low in primary aldosteronism because volume expansion suppresses it.

· Aldosterone: high in stenosis because angiotensin II stimulates the adrenal cortex; high in aldosteronism because secretion is autonomous.

· Potassium and acid-base: both can give hypokalemia with metabolic alkalosis, because in each case aldosterone acts on the collecting duct.

· Clues: stenosis gives an abdominal bruit, a small kidney on the affected side, and a string of beads appearance in fibromuscular disease; aldosteronism gives hypokalemia with no bruit and normal kidneys, often found during work-up of resistant hypertension.

· Edema: neither develops edema. In aldosteronism, rising pressure and atrial natriuretic peptide restore sodium balance within days, which is aldosterone escape.

· Blocking angiotensin II: in stenosis the efferent arteriole dilates and the glomerular filtration rate of that kidney falls sharply, and bilateral disease can cause a large fall; in aldosteronism angiotensin II is not driving the aldosterone, so blocking it does not correct the abnormality.

Distinctions that matter

· The renin level is the single discriminating cell in this table; aldosterone is high in both.

· A high renin state is driven from the kidney, and a low renin state with high aldosterone is driven from the adrenal gland.

Common misconceptions

· Writing that primary aldosteronism causes edema because sodium is retained. Escape resets sodium balance and the extra sodium shows as pressure, not swelling.

· Treating a falling glomerular filtration rate after angiotensin II blockade as a drug toxicity unrelated to anatomy. It reveals that filtration depended on efferent constriction in a stenotic kidney.

Where it is taught: Section C02, renal artery stenosis and adrenal causes paragraphs with the secondary causes table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C03-BD1

Brain dump 1 · foundational retrieval: Hyaline arterioles and the concentrically hypertrophied ventricle

A competent response contains

· Wall stress is proportional to pressure times radius divided by twice the wall thickness, which is the law of Laplace.

· A higher afterload raises systolic wall stress, and myocytes respond by adding sarcomeres in parallel, so the wall thickens and the denominator of that relationship rises, returning stress toward normal.

· Pressure overload gives concentric hypertrophy, a thick wall with little chamber enlargement; volume overload adds sarcomeres in series and dilates the chamber, which is eccentric hypertrophy.

· The normal left ventricular wall is about 1.0 to 1.5 cm and can approach or exceed 2 cm in hypertensive heart disease.

· Hypertrophied myocytes are thicker with enlarged rectangular boxcar nuclei.

· Hyaline arteriolosclerosis: chronic stress injures endothelium, plasma proteins leak into the wall, and smooth muscle cells add basement membrane and matrix.

· The result is a homogeneous, pink, glassy wall with a narrowed lumen, causing slow patchy ischemia of the tissue supplied.

· Arterioles are the resistance vessels, which is why they take the strain.

· The same hyaline lesion is typical of diabetes and appears mildly with normal aging.

Distinctions that matter

· Concentric against eccentric hypertrophy follows from the load: pressure adds sarcomeres in parallel, volume adds them in series.

· Hypertrophy is an adaptation that normalizes systolic wall stress, and the cost appears in diastole.

Common misconceptions

· Saying hypertrophy raises wall stress. Thickening is the term in the denominator, so it lowers stress back toward normal.

· Treating hyaline arteriolosclerosis as specific to hypertension. It also occurs in diabetes and mildly with aging.

Where it is taught: Section C03, hyaline arteriolosclerosis and concentric left ventricular hypertrophy of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C03-BD2

Brain dump 2 · integration and application: Trace chronic pressure into four organs

A competent response contains

· A thick, fibrotic ventricle relaxes slowly and is stiff, so a normal end-diastolic volume requires a higher filling pressure; left atrial and pulmonary venous pressures rise and the lungs congest although the ejection fraction is normal.

· Atrial fibrillation removes atrial contraction, which this stiff ventricle depends on for filling, and the fast rate shortens diastole, so filling falls further.

· Atrial contraction into a stiff ventricle produces a fourth heart sound.

· ECG: tall R waves in V5 and V6 with deep S waves in V1 reflect the greater mass, and lateral ST depression with T wave inversion forms the strain pattern.

· Ischemia in a hypertrophied wall: greater muscle mass, capillary density that lags behind the muscle, and high diastolic pressure compressing subendocardial vessels, all worsened by accelerated coronary atherosclerosis.

· Kidney: benign nephrosclerosis, with hyaline afferent arterioles and thickened interlobular arteries causing glomerulosclerosis, tubular atrophy, and interstitial fibrosis; the kidneys are symmetrically small with a finely granular surface, proteinuria is mild, and glomerular filtration rate falls slowly.

· Hypertension is the second leading cause of end-stage kidney disease after diabetes, with higher risk in carriers of high-risk APOL1 variants, and nephron loss worsens the hypertension by impairing sodium excretion.

· Retina: arteriolar narrowing, copper or silver wiring, arteriovenous nicking, flame hemorrhages, hard exudates, and cotton wool spots, which are nerve fiber layer microinfarcts.

· Arteriovenous nicking occurs because a thickened arteriole compresses a vein where the two share a connective tissue sheath.

· Brain: lipohyalinosis occludes penetrating arteries and causes lacunar infarcts under about 15 mm in the basal ganglia, internal capsule, thalamus, and pons; Charcot-Bouchard microaneurysms weaken the same vessels and rupture causes deep intracerebral hemorrhage, most often in the putamen, then thalamus, pons, and cerebellum.

· The bleeds sit deep because penetrating arteries such as the lenticulostriate branches of the middle cerebral artery receive nearly full parent arterial pressure; amyloid angiopathy affects superficial cortical and leptomeningeal vessels, so its hemorrhages are lobar.

Distinctions that matter

· Heart failure with a preserved ejection fraction is a filling problem, so the ejection fraction cannot exclude it.

· Lacunar infarcts and deep hemorrhages share one vascular bed and two different consequences of the same arteriolar disease.

Common misconceptions

· Expecting a normal ejection fraction to rule out pulmonary congestion. The limiting problem here is diastolic filling pressure.

· Explaining subendocardial ischemia only by coronary atherosclerosis. Mass, capillary supply, and diastolic compression contribute independently.

Where it is taught: Section C03, the paragraph after the pressure overload flow figure and the kidney, retina, and brain paragraphs, with the stop and predict box of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C03-RT01

Retrieval exercise: Rebuild the pressure overload sequence

A competent response contains

· Step 1: sustained hypertension raises left ventricular afterload.

· Step 2: systolic wall stress rises, by the law of Laplace, in which stress is proportional to pressure times radius divided by twice the wall thickness; pressure is the term that rises first.

· Step 3: sarcomeres are added in parallel, giving concentric hypertrophy, so the wall thickens while chamber volume changes little.

· Step 4: the thick, fibrotic wall relaxes slowly and is stiff.

· Step 5: a higher left atrial pressure is needed for filling, because a stiff ventricle needs a greater pressure to reach the same end-diastolic volume.

· Step 6: the left atrium dilates and pulmonary venous pressure rises.

· Step 7: atrial fibrillation and heart failure with preserved ejection fraction follow.

· First drawing: wall about 1.0 to 1.5 cm with a normally sized atrium. Final drawing: wall approaching or above 2 cm, a cavity of similar or slightly smaller size, and a dilated left atrium.

· Very late in the course the ventricle can dilate and lose systolic function, so the ejection fraction falls.

Distinctions that matter

· Hypertrophy succeeds in systole and fails in diastole, which is why the ejection fraction can stay normal while symptoms appear.

· Atrial dilation is a consequence of the filling pressure, which is why atrial fibrillation belongs at the end of this sequence rather than as a separate problem.

Common misconceptions

· Drawing a large dilated chamber for pressure overload. The chamber changes little; dilation belongs to volume overload and to the late failing stage.

· Placing atrial fibrillation before atrial dilation. The raised filling pressure stretches the atrium first.

Where it is taught: Section C03, flow figure on pressure overload to diastolic failure and atrial fibrillation of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C04-BD1

Brain dump 1 · foundational retrieval: Autoregulation, breakthrough, and the acute arteriolar lesion

A competent response contains

· A hypertensive emergency is severe elevation, often 180/120 mmHg or higher, with new or worsening acute damage to a target organ.

· Equally high pressure without acute organ damage is sometimes called hypertensive urgency.

· Malignant hypertension was the older term for severe hypertension with retinal hemorrhages, exudates, or papilledema, often with acute kidney injury, and is now treated as a form of emergency.

· The rate of rise and the failure of protective mechanisms matter more than the level reached.

· The brain and kidney hold flow nearly constant across a range of pressures by arteriolar constriction and dilation; in the brain this range is roughly 60 to 150 mmHg mean arterial pressure.

· Chronic hypertension shifts the autoregulation curve to the right, so both the lower and the upper limits move up.

· Above the upper limit, arteriolar constriction is forced open, which is breakthrough; capillaries then meet full arterial pressure and the blood-brain barrier leaks.

· The result is vasogenic edema and hypertensive encephalopathy with headache, confusion, visual disturbance, and seizures.

· The edema often favors the posterior circulation, which has less sympathetic innervation, giving posterior reversible encephalopathy syndrome.

· Fibrinoid necrosis is necrosis of the vessel wall into which fibrin and other plasma proteins seep, appearing bright pink and smudgy on routine stains, often with luminal thrombus.

· Surviving arterioles build concentric layers of smooth muscle with reduplicated basement membrane, the onion-skin lesion of hyperplastic arteriolosclerosis, most conspicuous in renal interlobular arteries and afferent arterioles.

· The same lesion occurs in scleroderma renal crisis and other thrombotic microangiopathies.

Distinctions that matter

· Emergency against urgency is decided by acute target organ damage, not by the number on the cuff.

· Hyaline thickening is the chronic lesion and onion-skin hyperplasia with fibrinoid necrosis is the acute severe lesion.

Common misconceptions

· Using a pressure threshold alone to call an emergency. Without acute organ injury the same pressure is urgency.

· Describing autoregulation as protective at any pressure. Beyond the upper limit constriction is overcome and the protection is lost.

Where it is taught: Section C04, opening paragraph, autoregulation and breakthrough, and the acute arteriolar lesion of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C04-BD2

Brain dump 2 · integration and application: Why the lower pressure did more damage

A competent response contains

· The pregnant woman has a hypertensive emergency; the older woman has severe hypertension without acute organ damage, which is urgency.

· The older woman's autoregulation curve has shifted right over 30 years, so 186/112 mmHg still lies within her range. The previously normotensive woman has an unshifted curve, so 168/108 mmHg exceeds her upper limit, causing breakthrough, vasogenic edema, and encephalopathy.

· Rising creatinine and hematuria come from fibrinoid necrosis and onion-skin narrowing of renal arterioles with glomerular ischemia, which is acute kidney injury.

· Fibrin strands in damaged small vessels shear red cells into schistocytes, giving microangiopathic hemolytic anemia with thrombocytopenia from platelet consumption and a raised lactate dehydrogenase from hemolysis.

· Renal vicious cycle: severe rise in arterial pressure; fibrinoid necrosis and onion-skin narrowing of renal arterioles; glomerular ischemia downstream; juxtaglomerular cells release large amounts of renin; angiotensin II constricts arterioles while pressure natriuresis shrinks plasma volume and adds further renin; pressure rises further and injures more arterioles.

· Renin is very high here because arteriolar injury makes the glomeruli ischemic, which is a direct stimulus to the juxtaglomerular cells, whereas in chronic hypertension renin is variable and often normal or low.

· Pressure natriuresis at a very high pressure contracts plasma volume, and a smaller volume is itself a stimulus to renin, so the two stimuli add.

· Triggers: untreated primary hypertension, renovascular disease, glomerulonephritis, scleroderma renal crisis, preeclampsia, pheochromocytoma, cocaine or amphetamines, and abrupt clonidine withdrawal.

· Abrupt clonidine withdrawal belongs on the list because it causes a rebound sympathetic surge.

· Other acute injuries follow the same logic: sudden afterload can cause left ventricular failure or ischemia, and shear stress can tear the aortic intima, causing dissection.

Distinctions that matter

· The position of the autoregulation curve, not the absolute pressure, decides who breaks through.

· Acute severe hypertension raises renin through renal ischemia and volume contraction, which is the opposite direction from the volume expansion of most chronic hypertension.

Common misconceptions

· Ranking severity by the higher cuff reading. The chronically hypertensive woman tolerates a higher pressure because her curve has shifted.

· Expecting plasma volume to be expanded in a hypertensive emergency. Pressure natriuresis contracts it and that drives renin higher.

Where it is taught: Section C04, autoregulation and breakthrough, the renal vicious cycle figure, the comparison table, and the stop and predict box of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C04-PR01

Prediction exercise: Predict what a rapid severe rise in pressure does

A competent response contains

· Cerebral arteriolar constriction is overcome and the arterioles are forced open, which is breakthrough of autoregulation; a previously normotensive person has an unshifted curve and breaks through at a lower pressure.

· Barrier permeability rises and brain water content rises, giving vasogenic edema, because capillaries now meet full arterial pressure.

· Renal arterioles show fibrinoid necrosis, bright pink and smudgy with fibrin in the wall, often with luminal thrombus, and surviving arterioles show onion-skin hyperplastic layering.

· Renin release rises sharply, because arteriolar narrowing makes the glomeruli ischemic and pressure natriuresis contracts plasma volume, and both stimulate the juxtaglomerular cells.

· The platelet count falls, the blood film shows schistocytes, and lactate dehydrogenase rises, because fibrin strands in injured small vessels shear red cells and consume platelets.

· The kidney surface shows petechial hemorrhages, the flea-bitten appearance, rather than the symmetrically small, finely granular kidneys of long-standing moderate hypertension.

· The left ventricle has had no time to hypertrophy, so the sudden afterload can cause acute failure with pulmonary edema, or ischemia.

Distinctions that matter

· Vasogenic edema follows barrier leak under pressure, which is not the same as the cytotoxic swelling of ischemic cells.

· Flea-bitten against finely granular kidney surface separates the acute severe lesion from the chronic one.

Common misconceptions

· Expecting the kidneys to be small and granular in an acute emergency. That appearance takes years of benign nephrosclerosis.

· Treating the low platelet count as a separate bleeding disorder. It is consumption within injured small vessels.

Where it is taught: Section C04, autoregulation and breakthrough, the acute arteriolar lesion, the comparison table, and the renal vicious cycle figure of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C05-BD1

Brain dump 1 · foundational retrieval: Pulmonary pressures, the resistance equation, and the five groups

A competent response contains

· Normal pulmonary artery pressure is about 25/8 mmHg with a mean of about 14 mmHg, and pulmonary vascular resistance is about one tenth of systemic resistance or less.

· During exercise, recruitment and distension of capillaries lower resistance as flow rises.

· The thin-walled right ventricle handles a volume load well but tolerates a rise in pressure poorly.

· Pulmonary hypertension is a mean pulmonary artery pressure above 20 mmHg at rest on right heart catheterization, the 2018 World Symposium threshold; many textbooks still quote 25 mmHg.

· Echocardiography raises the suspicion first, estimating systolic pressure from the tricuspid regurgitant jet.

· Mean pulmonary artery pressure equals cardiac output times pulmonary vascular resistance plus left atrial pressure, which is estimated by the wedge pressure.

· Pulmonary vascular resistance is mean pulmonary artery pressure minus wedge pressure, divided by cardiac output, in Wood units; one Wood unit is 1 mmHg per L/min.

· Precapillary: wedge pressure 15 mmHg or less with resistance above 2 Wood units. Isolated postcapillary: wedge pressure above 15 mmHg with resistance of 2 or less. Combined: high wedge pressure and high resistance.

· Group 1, pulmonary arterial hypertension: disease of small pulmonary arteries with endothelial dysfunction, constriction, and proliferative remodeling; idiopathic, heritable through BMPR2, anorexigens and methamphetamine, systemic sclerosis, HIV, portal hypertension, congenital shunts, schistosomiasis.

· Group 2, left heart disease: high left atrial pressure transmitted backward; heart failure with reduced or preserved ejection fraction, mitral and aortic valve disease. This is the most common cause.

· Group 3, lung disease or hypoxia: hypoxic vasoconstriction, capillary loss, remodeling; chronic obstructive pulmonary disease, interstitial lung disease, sleep apnea, hypoventilation, high altitude. This is the next most common.

· Group 4, pulmonary artery obstruction: organized thrombi obstruct arteries and open vessels remodel; chronic thromboembolic pulmonary hypertension.

· Group 5, unclear or multifactorial: sarcoidosis, myeloproliferative disorders.

Distinctions that matter

· Mean pressure alone does not localize the disease; the wedge pressure and the calculated resistance do.

· Group 1 is uncommon but carries the molecular pathogenesis, while groups 2 and 3 carry the numbers of patients.

Common misconceptions

· Treating an echocardiographic estimate as the diagnosis. The definition rests on right heart catheterization.

· Assuming a high mean pressure means disease of the pulmonary arteries. A high wedge pressure with normal resistance places the problem behind the left atrium.

Where it is taught: Section C05, opening paragraph, definition and hemodynamic types, and the classification table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C05-BD2

Brain dump 2 · integration and application: Separate two catheter studies and explain two mechanisms

A competent response contains

· Patient A: (36 minus 10) divided by 5 equals 5.2 Wood units with a normal wedge pressure, so this is precapillary disease; examples are pulmonary arterial hypertension, chronic lung disease, and chronic thromboembolic disease.

· Patient B: (36 minus 28) divided by 5 equals 1.6 Wood units, which is normal, with a high wedge pressure, so this is isolated postcapillary disease; examples are left ventricular failure and mitral stenosis.

· The same mean pressure can be produced by resistance within the lung or by pressure transmitted backward from the left atrium, so the wedge pressure and resistance decide which organ is at fault.

· Hypoxic vasoconstriction in order: low alveolar oxygen closes oxygen-sensitive potassium channels in pulmonary arterial smooth muscle; the cells depolarize; L-type calcium channels open; calcium enters; the muscle contracts.

· Locally this diverts blood away from poorly ventilated alveoli and improves the matching of ventilation and perfusion.

· When the whole lung is hypoxic the response constricts the entire pulmonary bed, raising resistance and driving medial hypertrophy.

· Chronic hypoxemia drives polycythemia, and the higher viscosity raises resistance further.

· In group 2, raised left atrial pressure first raises pulmonary pressure passively, but persistent venous hypertension, classically from long-standing mitral stenosis, remodels the arteries and adds a precapillary component, so the resistance rises as well.

· In group 4 a small minority of emboli organize into fibrous webs instead of lysing, and ventilation-perfusion scanning shows mismatched perfusion defects, whereas group 1 shows no such mismatched defects.

Distinctions that matter

· Systemic arterioles dilate with hypoxia and pulmonary arterioles constrict, which is why lung disease raises only pulmonary resistance.

· A combined picture is a postcapillary disease that has acquired a precapillary component, so both the wedge pressure and the resistance are high.

Common misconceptions

· Calculating resistance from mean pressure divided by cardiac output alone. The wedge pressure must be subtracted first.

· Assuming hypoxic vasoconstriction is always useful. It helps regionally and harms globally.

Where it is taught: Section C05, worked example on precapillary and postcapillary disease and the paragraph after the classification table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C06-BD1

Brain dump 1 · foundational retrieval: Rebuild the pathogenesis of pulmonary arterial hypertension

A competent response contains

· BMPR2 is the gene most often mutated in heritable pulmonary arterial hypertension; it is a receptor of the transforming growth factor beta superfamily, signals through SMAD proteins, and normally restrains smooth muscle proliferation and promotes apoptosis.

· Inheritance is autosomal dominant with incomplete penetrance, so a second hit such as inflammation or hypoxia is thought to be needed.

· Nitric oxide acts through soluble guanylyl cyclase and cGMP, dilating and acting against proliferation, and is decreased.

· Prostacyclin acts through the IP receptor, Gs, and cAMP, dilating and inhibiting platelets, and is decreased while thromboxane A2 rises.

· Endothelin-1 acts through the ETA receptor, Gq, and calcium, constricting and acting as a mitogen, and is increased.

· Sequence: BMPR2 loss or another trigger injures the endothelium; nitric oxide and prostacyclin fall while endothelin-1 and thromboxane rise; small arteries constrict and in situ thrombi form; apoptosis-resistant smooth muscle proliferates; medial hypertrophy, intimal fibrosis, and plexiform lesions develop; pulmonary vascular resistance becomes fixed; the right ventricle hypertrophies and then fails, which is cor pulmonale.

· Structural order: medial hypertrophy with muscularization of normally nonmuscular arterioles first, then concentric intimal fibrosis.

· Plexiform lesions are tufts of slit-like endothelial channels at small artery branch points that resemble a glomerulus, seen in severe idiopathic, heritable, and shunt-related disease.

· Large pulmonary arteries develop atherosclerotic plaques, which do not normally form at pulmonary pressures.

· Acquired triggers and associations: serotonergic anorexigens such as fenfluramine, since serotonin is a smooth muscle mitogen, methamphetamine, systemic sclerosis, HIV, portal hypertension, congenital shunts, and hepatosplenic Schistosoma mansoni infection, in which eggs reach the lungs through portosystemic collaterals and provoke granulomatous arteriolar injury.

· Idiopathic disease is more common in women.

· A large left-to-right shunt remodels the arteries until pulmonary resistance exceeds systemic resistance and the shunt reverses, which is Eisenmenger syndrome.

Distinctions that matter

· Two dilator and antiproliferative signals fall while one constrictor and mitogen rises, so the imbalance acts on both tone and growth.

· Constriction and thrombosis are reversible features, while proliferative remodeling is what makes the resistance fixed.

Common misconceptions

· Treating BMPR2 loss as sufficient on its own. Penetrance is incomplete and a second hit is thought to be required.

· Calling pulmonary arterial hypertension a vasoconstrictive disease only. The remodeling, including plexiform lesions, is the reason resistance stops responding.

Where it is taught: Section C06, genetic and acquired triggers, the endothelial mediator table, and the pathogenesis flow figure of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C06-BD2

Brain dump 2 · integration and application: Explain the signs of the pressure-loaded right heart

A competent response contains

· Loud P2: high pulmonary artery diastolic pressure closes the pulmonic valve forcefully.

· Left parasternal heave and right-sided fourth heart sound: a hypertrophied, stiff right ventricle.

· Holosystolic murmur louder on inspiration with large v waves: tricuspid regurgitation from annular dilation.

· Raised jugular venous pressure, hepatomegaly, and edema: raised systemic venous pressure.

· Right axis deviation, tall R wave in V1, and P pulmonale: right ventricular mass and right atrial enlargement.

· The right ventricle first hypertrophies concentrically, then dilates.

· Dilation stretches the tricuspid annulus, so the valve leaks, and pushes the interventricular septum to the left into a D shape, which limits left ventricular filling.

· When right ventricular pressure nears systemic pressure, its normal systolic coronary perfusion is lost while demand rises, so ischemia accelerates the failure.

· Cor pulmonale is right ventricular hypertrophy, dilation, or failure caused by disease of the lungs or pulmonary vessels, excluding left heart disease and congenital heart disease.

· It is chronic and hypertrophic in chronic obstructive pulmonary disease or pulmonary arterial hypertension, and acute and dilated after a massive pulmonary embolism.

· Systemic pressure can be low because the right ventricle cannot push enough blood across the high resistance, so left atrial return falls, and the shifted septum restricts left ventricular filling; low preload lowers stroke volume.

· Exertional syncope means cardiac output cannot rise across a fixed resistance, so it marks advanced disease.

· A dilated pulmonary artery can compress the left recurrent laryngeal nerve, causing hoarseness.

Distinctions that matter

· Hypertrophy gives the heave and the fourth heart sound, while dilation gives the regurgitant murmur and the venous congestion, so the findings date the stage.

· The left ventricle underfills here rather than failing, which is why systemic pressure falls with normal left ventricular structure.

Common misconceptions

· Reading a low systemic pressure as left ventricular disease. The limitation is right ventricular output and restricted left ventricular filling.

· Treating cor pulmonale as any right ventricular failure. It excludes failure caused by left heart disease and by congenital heart disease.

Where it is taught: Section C06, the right ventricle and cor pulmonale, the table of findings and mechanisms, and the stop and predict box of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C06-CA01

Clinical application: Progressive breathlessness in a young woman with a loud P2

A competent response contains

· Recognize: resistance is (52 minus 9) divided by 3.4, which is about 12.6 Wood units, with a wedge pressure of 9 mmHg, so this is precapillary pulmonary hypertension; with no mismatched perfusion defects and no lung or connective tissue disease, she belongs to group 1, pulmonary arterial hypertension, and the family history points to the heritable form.

· Explain: a loss-of-function BMPR2 mutation removes a transforming growth factor beta superfamily brake that normally restrains smooth muscle proliferation and promotes apoptosis through SMAD signaling; endothelial injury then lowers nitric oxide and prostacyclin and raises endothelin-1 and thromboxane, so small arteries constrict and thrombose in situ, apoptosis-resistant smooth muscle proliferates, and medial hypertrophy, intimal fibrosis, and plexiform lesions fix the resistance.

· Differentiate: left heart disease is excluded by the wedge pressure of 9 mmHg, which shows left atrial pressure is normal; chronic lung disease is excluded by clear lungs, no smoking history, and no hypoxic lung disease; chronic thromboembolic disease is excluded by the absence of mismatched perfusion defects on ventilation-perfusion scanning.

· Investigate: a family history and genetic testing for BMPR2 for the heritable form; skin tightening, Raynaud phenomenon, and connective tissue serology for systemic sclerosis; HIV testing; signs of portal hypertension and liver disease; travel or residence history for schistosomiasis; an anorexigen or methamphetamine exposure history; echocardiography for a congenital shunt, right ventricular hypertrophy and dilation, septal flattening, and tricuspid regurgitation.

· Decide: the right ventricle is failing; the raised jugular venous pressure with large v waves, the hepatomegaly, the ankle edema, and the inspiratory holosystolic murmur of tricuspid regurgitation show failure and dilation rather than hypertrophy alone; right ventricular failure from pulmonary vascular disease is cor pulmonale.

· Justify: output is low because the right ventricle cannot push blood across a fixed resistance of about 12.6 Wood units and because its dilation shifts the septum left and restricts left ventricular filling, so left ventricular preload and stroke volume fall despite normal left ventricular structure; she fainted on exertion because output cannot rise across that fixed resistance, so cerebral perfusion falls with exercise; she is hoarse because a dilated pulmonary artery compresses the left recurrent laryngeal nerve.

Distinctions that matter

· A normal wedge pressure with a high resistance localizes the disease to the pulmonary arteries, which is the single most useful line in the catheter report.

· Exertional syncope and a falling cardiac output reflect a fixed resistance, not a failing left ventricle.

Common misconceptions

· Reading a low systemic pressure in severe pulmonary arterial hypertension as left ventricular failure.

· Assuming any pulmonary hypertension with right heart failure is chronic thromboembolic disease. The perfusion scan separates them.

Where it is taught: Section C05, definition and hemodynamic types with the worked example, and Section C06, pathogenesis and the right ventricle and cor pulmonale of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C04-Q01

Board-style question: Arteriolar lesion in an abrupt severe rise in pressure

Correct answer: C. Concentric laminated layers of smooth muscle with fibrin deposition and wall necrosis

Why it is right: A rapid, severe rise injures arteriolar endothelium, so fibrin and plasma proteins enter the wall and smooth muscle cells die, giving fibrinoid necrosis, while surviving arterioles lay down concentric layers of smooth muscle with reduplicated basement membrane, the onion-skin lesion of hyperplastic arteriolosclerosis.

A is wrong: Homogeneous glassy pink thickening with a narrowed lumen is hyaline arteriolosclerosis, the lesion of years of moderate hypertension, diabetes, and aging, and it does not explain acute kidney injury 4 months after a normal reading.

B is wrong: Subendothelial lipid-laden macrophages under a fibrous cap describe an atherosclerotic plaque, which forms in large and medium arteries rather than in arterioles.

D is wrong: Nodular mesangial matrix expansion with thickened capillary basement membranes is diabetic glomerular disease, which involves the glomerulus rather than the arteriolar wall and develops over years.

A competent response contains

· A previously normotensive patient whose pressure rises abruptly to 212/128 mmHg with encephalopathy, papilledema, acute kidney injury, thrombocytopenia, and schistocytes has a hypertensive emergency.

· The arteriolar lesion of a rapid severe rise is fibrinoid necrosis with concentric onion-skin hyperplastic arteriolosclerosis, most conspicuous in renal interlobular arteries and afferent arterioles.

Distinctions that matter

· Onion-skin layering with fibrinoid necrosis marks acute severe hypertension, while glassy hyaline thickening marks chronic moderate hypertension.

Common misconceptions

· Choosing the chronic hyaline lesion because the patient is hypertensive. The time course and the acute organ injury point to the hyperplastic lesion.

Where it is taught: Section C04 of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C05-Q02

Board-style question: Reading a right heart catheter study

Correct answer: A. Backward transmission of a raised left atrial pressure

Why it is right: A wedge pressure of 26 mmHg with a calculated resistance of 2 Wood units shows that raised left atrial pressure is being transmitted backward, which is group 2 pulmonary hypertension from left heart disease, and her concentric hypertrophy with diastolic dysfunction and left atrial enlargement is the source of that pressure.

B is wrong: Proliferative remodeling with plexiform lesions is group 1 disease, which gives a wedge pressure of 15 mmHg or less with a resistance above 2 Wood units, the opposite of this study.

C is wrong: Hypoxic vasoconstriction with loss of alveolar capillaries is group 3 disease, which is precapillary and requires chronic lung disease or hypoxia, neither of which is present.

D is wrong: Organized thrombus obstructing the pulmonary arteries is group 4 disease, which is also precapillary and shows mismatched perfusion defects rather than a high wedge pressure.

A competent response contains

· Resistance here is (34 minus 26) divided by 4, which is 2 Wood units, a normal value, with a wedge pressure of 26 mmHg.

· A high wedge pressure with normal resistance is isolated postcapillary pulmonary hypertension, and the thick left ventricle with a preserved ejection fraction and a dilated left atrium identifies hypertensive heart disease as the cause.

Distinctions that matter

· The wedge pressure separates disease behind the left atrium from disease within the pulmonary arteries, whatever the mean pulmonary artery pressure is.

Common misconceptions

· Treating any mean pulmonary artery pressure above 20 mmHg as pulmonary arterial disease without calculating resistance.

Where it is taught: Section C05 of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.